

Task 2

Task Title:

Build an AI Agent for Dynamic Task Execution

Task Description:

Build an AI-powered agent that integrates a real Large Language Model (LLM) to dynamically interpret and execute user requests.

The agent should be capable of:

- Understanding the user's goal from natural language input
- Extracting structured data required to fulfill the request
- Selecting the appropriate tool based on the detected intent
- Performing real operations using external tools
- Generating a clear, well-structured final response

The system must operate entirely on real-time user input, without relying on predefined static scenarios.

Requirments:

1. Integrate a Real LLM

The system must connect to a real LLM (API or local model).

2. Use the LLM for:

- Intent recognition
- Structured parameter extraction (JSON format)
- Tool selection
- Final response generation

The LLM is the decision engine of the system.

Required Tools:

1. **Schedule Management Tool**

- Accept structured JSON input (date, time, event name)
- Create, update, or delete events
- Detect scheduling conflicts
- Return confirmation or conflict warnings

2. Location Info Tool

1. Take a location name as input
2. Return:

- Country
- Time zone
- Current local time (simulated or real)

Handle invalid or unknown locations

3. Simple Analytics Tool

1. Accept a list of numbers or values
2. Compute:

- Average
- Maximum / Minimum
- Count

Handle invalid or empty input gracefully